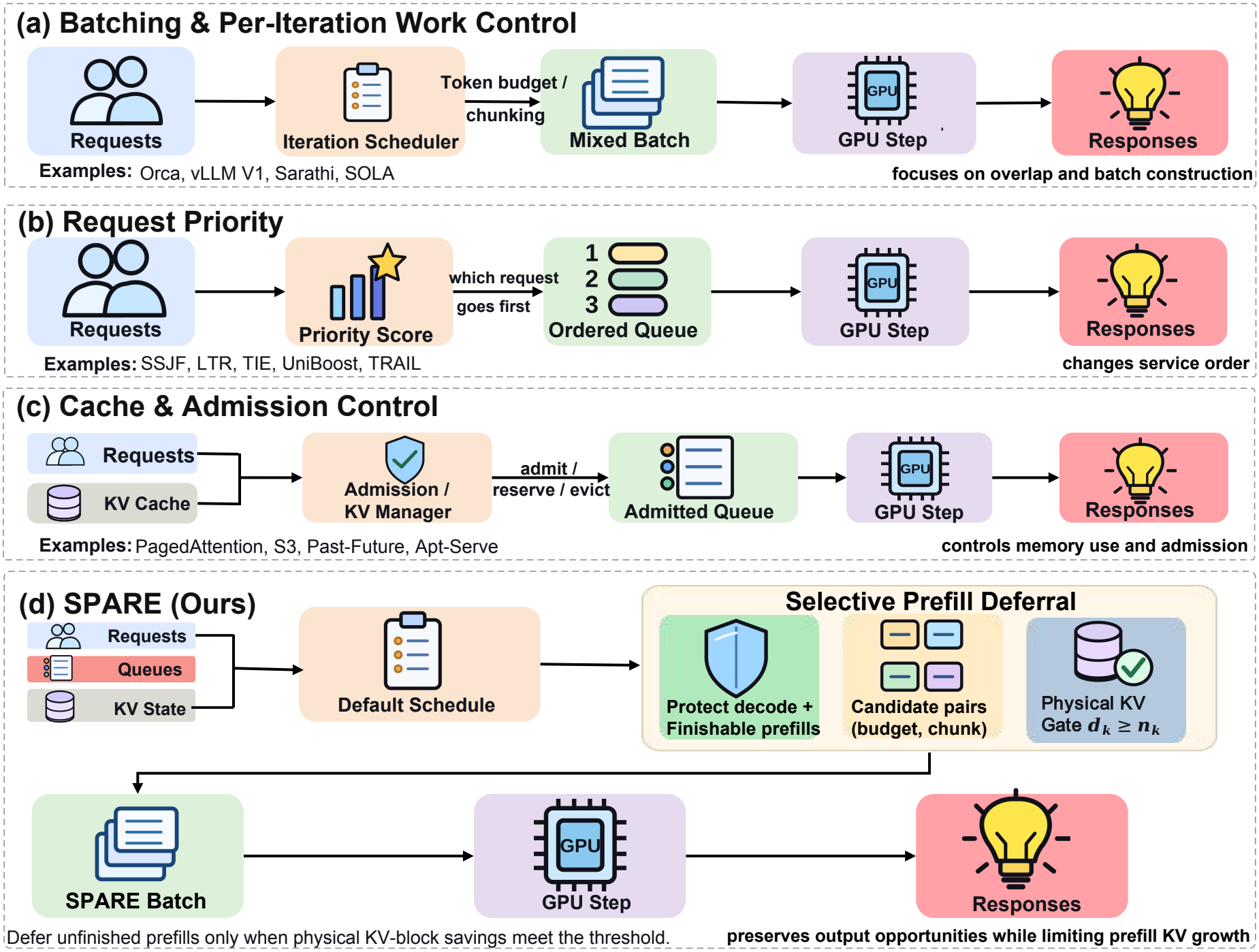

(a) Batching & Per-Iteration Work Control
Requests
Iteration Scheduler
Token budget / chunking
Mixed Batch
GPU
GPU Step
Responses
Examples: Orca, vLLM V1, Sarathi, SOLA
focuses on overlap and batch construction
(b) Request Priority
Requests
Priority Score
which request goes first
1
2
3
Ordered Queue
GPU
GPU Step
Responses
Examples: SSJF, LTR, TIE, UniBoost, TRAIL
changes service order
(c) Cache & Admission Control
Requests
KV Cache
Admission / KV Manager
admit / reserve / evict
Admitted Queue
GPU
GPU Step
Responses
Examples: PagedAttention, S3, Past-Future, Apt-Serve
controls memory use and admission
(d) SPARE (Ours)
Requests
Queues
KV State
Default Schedule
Selective Prefill Deferral
Protect decode + Finishable prefills
Candidate pairs (budget, chunk)
Physical KV Gate $d_k \geq n_k$
SPARE Batch
GPU
GPU Step
Responses
Defer unfinished prefills only when physical KV-block savings meet the threshold.
preserves output opportunities while limiting prefill KV growth